Global Aerospace

Tracking Engine Flight Hours

We conduct a detailed analysis of engine flight hour activity, which serves as a key indicator for assessing potential dynamics in aftermarket demand.

Global flight hours were 2.1% higher y/y in August (+1.5% y/y in July), reflecting the flight activity improvements in the Middle East. Narrow-body and wide-body flight hours saw an increase of 2.7% y/y and 0.5% y/y, respectively, during the month. NB aircraft represented 72% of total flight hours, led by CFM56 at 46%, LEAP at 29%, PW1000G at 12% and V2500 at 11% (Fig 39). WB aircraft accounted for 28% of flight hours, with GE90 engines at 24%, GEnx at 19%, Trent XWB at 15%, and Trent 700 at 10% (Fig 41).

Middle Eastern total flight hours recovery trend continued in August with total regional flight hours seeing a 1.9% y/y increase in August (compared to ~1% higher y/y in July, ~6% lower y/y in June, ~18% lower y/y in May, ~32% lower y/y in April and ~48% lower y/y in March). Narrow-body flight hours increased 1.7% y/y in August (vs +1% y/y in July, -7% y/y in June, -20% y/y in May, -38% y/y in April and -44% y/y in March) while wide-body utilization increased 2.0% y/y (vs +1% y/y in July, -5% y/y in June, ~17% y/y in May, ~27% y/y in April and ~50% y/y in March).

Melrose flight hours saw an increase of 0.6% y/y in August (vs +0.1% in July and -2.4% in June).

MTU flight hours increased 3.9% y/y in August (vs +3.0% in July and -0.9% in June). GTF activity rose 27% y/y, while V2500 declined 12% y/y.

Rolls-Royce flight hours grew 3.1% y/y in August (vs +2.8% in July and +2.8% in June). Trent XWB hours increased 9% y/y.

Safran flight hours increased 2.2% y/y in August (vs +1.5% in July and -1.0% in June). LEAP activity rose 21% y/y, while GE90 hours remained flat y/y. CFM56 activity saw a 7% y/y decline.

GE flight hours increased 1.6% y/y in August (vs +1.2% in July and -0.9% in June). Platform trends: LEAP +21%, GEnx +5%, GE90 flat, CFM56 -7% y/y.

Pratt & Whitney flight hours rose 5.6% y/y in August (vs +4.5% in July and +0.9% in June). GTF hours increased 27% y/y, while V2500 declined 12% y/y. PW2000 hours remained largely flat y/y.

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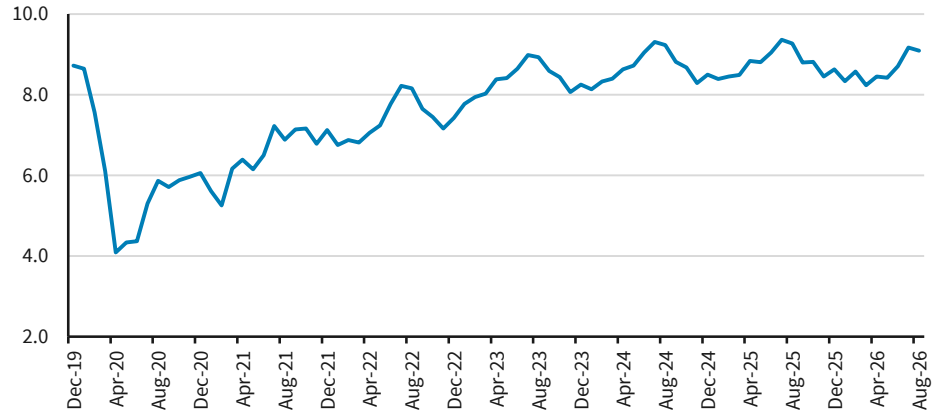
Please see analyst certifications and important disclosures beginning on page 26.
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Tracking the utilisation of the global engine fleet

We monitor engine flight activity across the global fleet of passenger and freighter aircraft. We track both flight hours and average daily utilisation by engine type. Total flight hours on an engine type incorporate both the number of engines in active service along with average utilisation. Average daily utilisation measures the daily usage of engines that are in service only (not parked for maintenance/storage).

The average daily utilisation of the global fleet stood at 9.1 hours in August, marginally lower than the ~9.2-9.3 hours flown historically during the peak summer months.

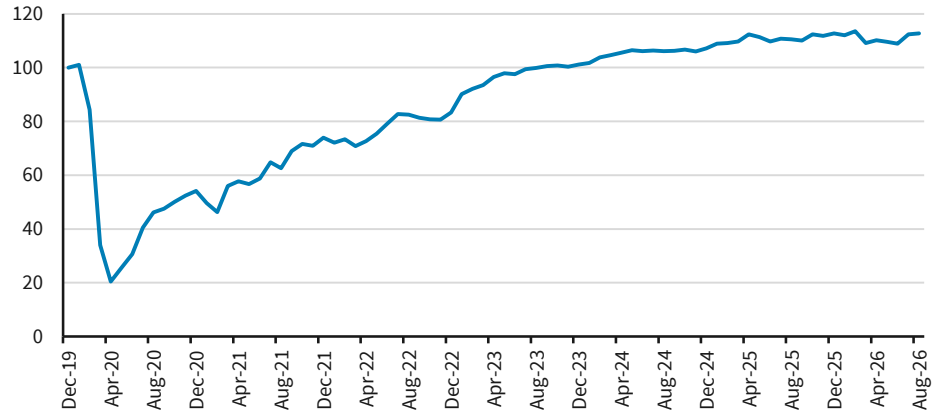
FIGURE 1. Average daily utilisation of the global NB and WB fleet (hours per day)



Source: Cirium, Barclays Research

We estimate total flight hours in the global narrow-body and wide-body fleet were 2.1% higher y/y in August, with the monthly flight hours being 12.7% above August 2019 levels.

FIGURE 2. Total flight hours of the global NB and WB fleet (indexed to 100 in the same month of 2019)

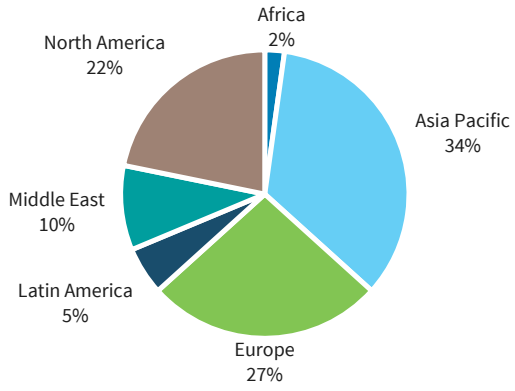


Source: Cirium, Barclays Research

Middle East flight hours continued to recover in August

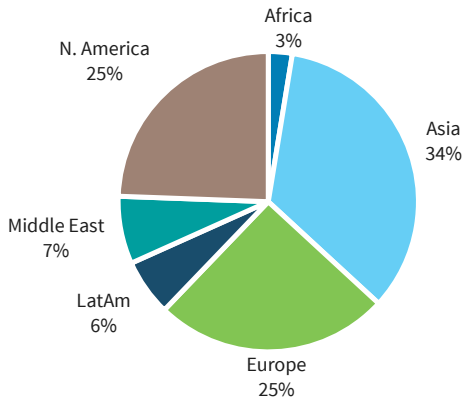
The Middle East is an important long-haul hub, accounting for ~7% of total flight activity hours in 2025 and a ~10% share of passenger RPKs.

FIGURE 3. The Middle East accounted for ~10% of the global passenger RPKs in 2025...



Source: IATA, Barclays Research

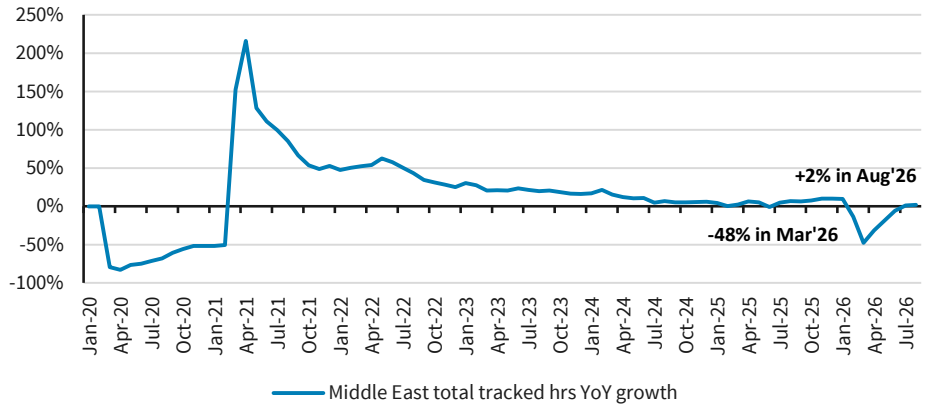
FIGURE 4. ... while the region's share of total tracked flight hours in 2025 stood at ~7%



Source: Cirium, Barclays Research

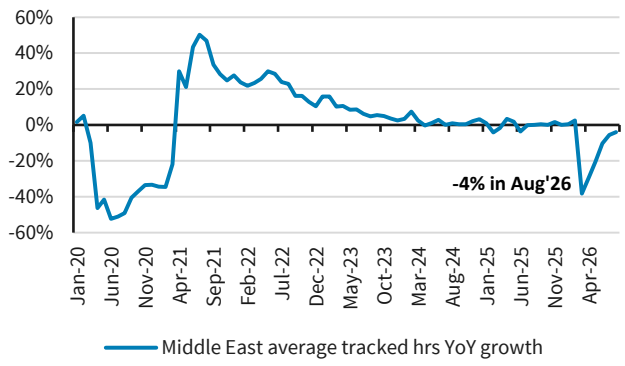
We observe that the total tracked flight hours in the Middle East saw a sharp 48% y/y decline in March before recovering to a 32% y/y decline in April, a 18% y/y decline in May and a 6% y/y decline in June. Flight activity in the region returned to growth in July with the monthly flight hours being 0.9% higher y/y. This **recovery trend continued in August** with the region's flight hours seeing a **1.9% y/y increase**. The total tracked hours are a function of the active fleet and the average utilisation level of the aircraft. Parking levels in the region represented ~19% of the installed fleet at end August (vs ~20% at end July and ~17% at the end of 2025). The average utilisation of aircraft in the active fleet saw a 4% decline y/y in August following a 6% decline y/y in July, 10% decline y/y in June, 21% decline y/y in May, a 29% decline y/y in April and a 38% decline y/y in March.

FIGURE 5. Total tracked flight hours in the Middle East continued to recover with +1.9% y/y growth in August - flight hours had seen a sharp 48% y/y decline in March following the start of the conflict



Source: Cirium, Barclays Research

FIGURE 6. Average fleet utilisation was ~4% lower y/y in August...



Source: Cirium, Barclays Research

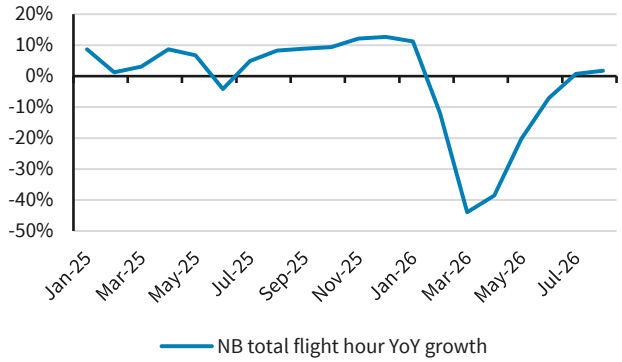
FIGURE 7. ... while parking levels in August represented ~19% of the installed fleet (vs ~17% at end 2025)



Source: Cirium, Barclays Research

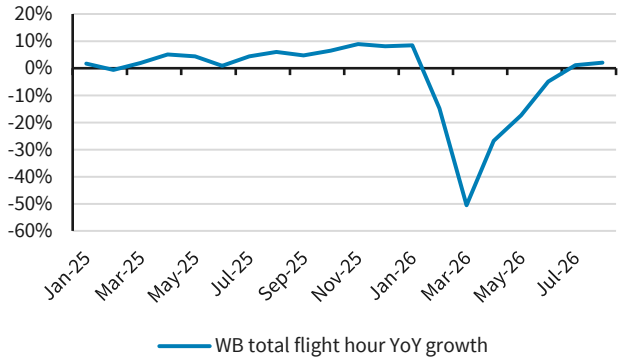
Total narrow-body flight hours in the Middle East saw an increase of +1.7% y/y in August (vs +0.7% y/y in July, -7.2% y/y in June, -20.3% y/y in May, -38.5% y/y in April and -43.9% y/y in March) while wide-body flight hours increased 2.0% y/y during the month (vs +1.1% y/y in July, -5.0% y/y in June, -17.4% y/y in May, -26.8% y/y in April and -50.5% y/y in March).

FIGURE 8. Narrow-body flight hours in the Middle East increased 1.7% y/y in August



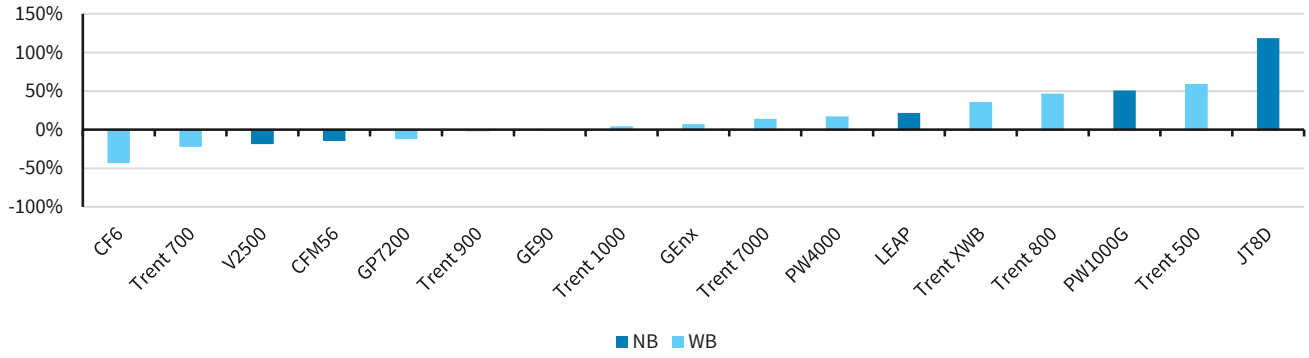
Source: Cirium, Barclays Research

FIGURE 9. Wide-body flight hours in the Middle East saw 2.0% y/y growth during the month



Source: Cirium, Barclays Research

FIGURE 10. YoY change in total flight hours of each engine type in the Middle East during August



Source: Cirium, Barclays Research

A look at the installed fleet exposure to the Middle East

Against the backdrop of the conflict, we assess each OEM's installed-fleet exposure to the Middle Eastern region. Given the region's role as a global long-haul hub, exposure is skewed to widebody platforms. Rolls-Royce shows the highest exposure, with ~9% of its installed fleet located in the region versus an industry average of ~7%, reflecting its widebody mix.

Beyond near-term demand volatility, the consequences for the global economy from higher oil price will likely depend on the duration of the conflict: historically, global air-traffic growth has tracked GDP at roughly 1.5x. In a weaker traffic environment, airlines have a limited set of levers to pull, and the most effective way to quickly adjust capacity is by parking or retiring older aircraft and by reducing utilisation rates. These decisions disproportionately affect the out-of-warranty fleet, which carries higher fuel and maintenance costs. And these same older aircraft are the primary drivers of aftermarket demand.

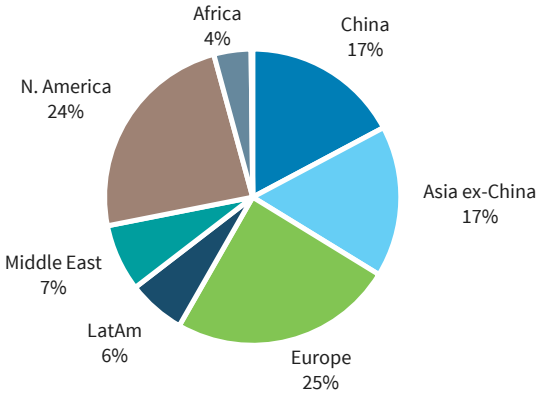
FIGURE 11. Installed fleet by engine type in the Middle East

Engine Type	Aircraft Type	ME aircraft count	ME engine count	Global engine count	*ME installed fleet share*
CF6	A300, A310, A330, 747, 767, DC-10, MD-11	68	166	3,008	5.5%
CFM56	A320ceo fly, A340, 737C, 737 NG, DC-8	438	912	21,238	4.3%
GE90	777	324	648	2,628	24.7%
GEnx	747, 787	157	314	2,308	13.6%
GP7200	A380	102	408	428	95.3%
JT8D	707, 727, 737C, DC-9, MD-80	64	134	778	17.2%
LEAP	A320neo fly, 737 MAX, C919	314	628	10,134	6.2%
PW1000G	A220, A320neo fly, E2 jets, MC-21	41	82	5,736	1.4%
PW2000	757	2	4	462	0.9%
PW4000	A300, A310, A330, 747, 767, 777, MD-11	27	78	1,405	5.6%
RB211	747, 757, 767	1	2	873	0.2%
Trent 500	A340	4	16	256	6.3%
Trent 700	A330ceo	53	106	1,510	7.0%
Trent 7000	A330neo	9	18	396	4.5%
Trent 800	777	14	28	324	8.6%
Trent 900	A380	33	132	380	34.7%
Trent 1000	787	28	56	846	6.6%
Trent XWB	A350	103	206	1,464	14.1%
V2500	A320ceo fly, MD-90	124	248	5,448	4.6%
Total		**1,906**	**4,186**	**59,840**	**7.0%**

Source: Cirium, Barclays Research

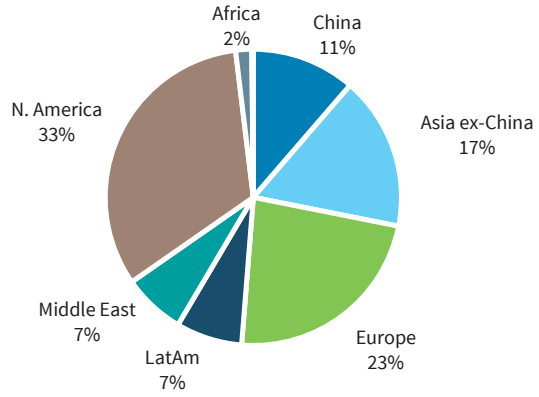
Regional split of the engine installed fleet

FIGURE 12. Regional split of Safran engine installed fleet



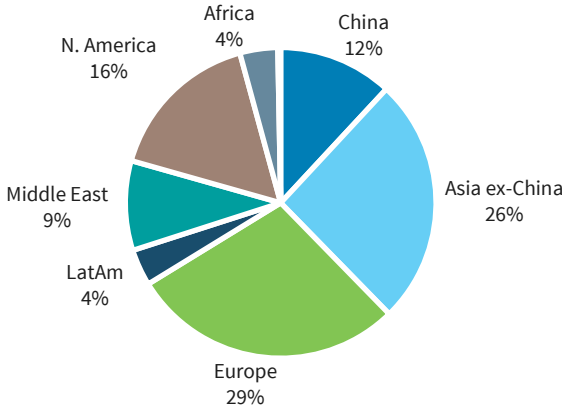
Source: Cirium, Barclays Research

FIGURE 13. Regional split of MTU engine installed fleet



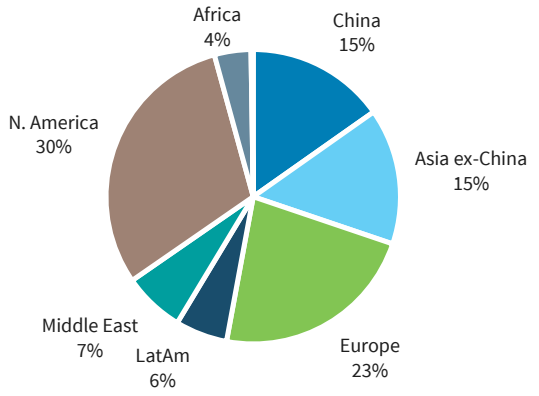
Source: Cirium, Barclays Research

FIGURE 14. Regional split of Rolls-Royce engine installed fleet



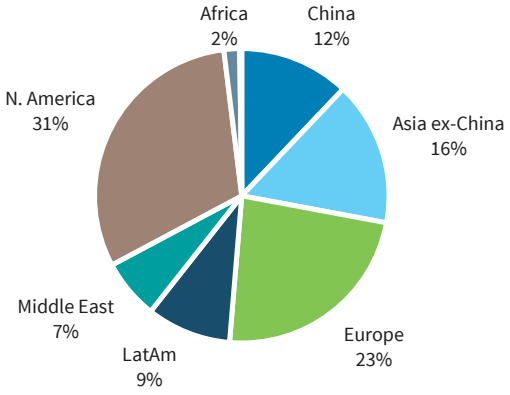
Source: Cirium, Barclays Research

FIGURE 15. Regional split of GE engine installed fleet



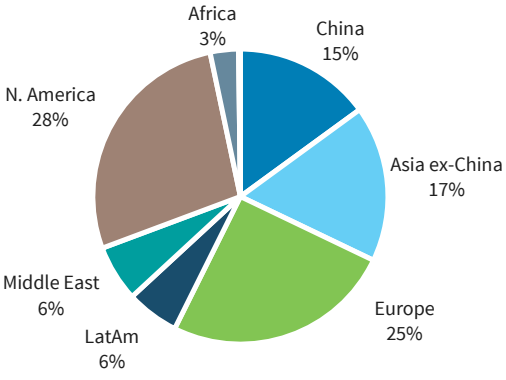
Source: Cirium, Barclays Research

FIGURE 16. Regional split of Pratt engine installed fleet



Source: Cirium, Barclays Research

FIGURE 17. Regional split of Melrose engine installed fleet

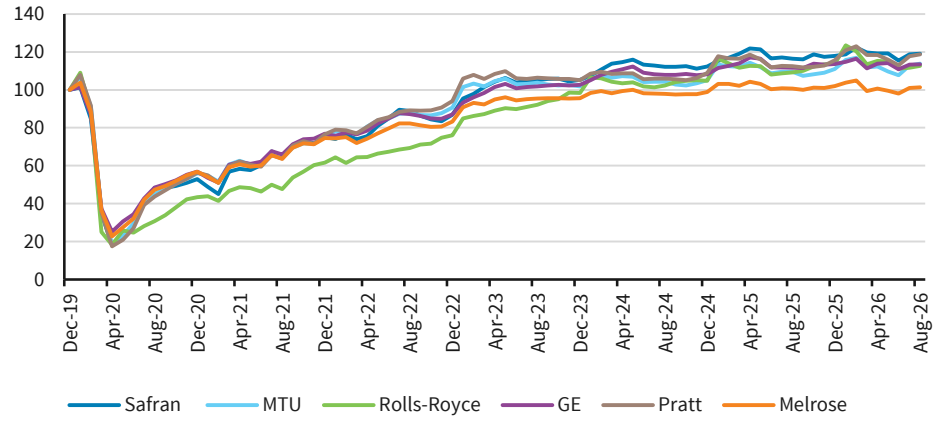


Source: Cirium, Barclays Research

Tracking flight hours by engine OEM

Based on our analysis, we estimate Safran total flight hours across its in-service engine fleet in August were 19.0% above August 2019 levels, while Rolls-Royce and MTU flight hours were 12.6% and 13.7% higher, respectively. Pratt and GE flight hours during the month were 18.8% and 13.5% higher vs 2019 levels, respectively.

FIGURE 18. Total flight hours by engine OEM (indexed to 100 in the same month of 2019)



Source: Cirium, Barclays Research

Safran flight hours increased 2.2% y/y in August, while MTU recorded a 3.9% y/y expansion. GE and P&W flight hours during the month increased 1.6% y/y and 5.6% y/y, respectively, while Melrose flight hours increased 0.6% y/y. Rolls-Royce engine activity hours were up 3.1% y/y, supported by a younger in-service fleet mix.

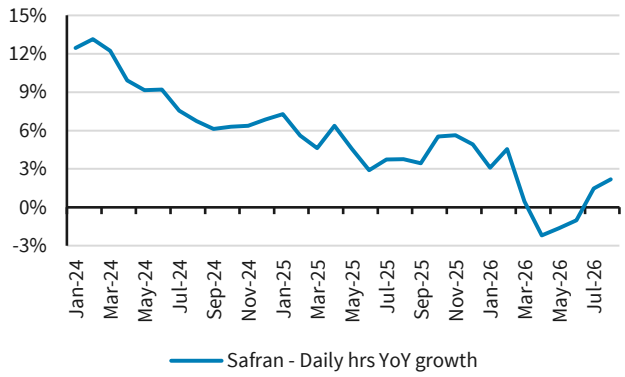
FIGURE 19. Flight hour growth by engine OEM

	YoY growth							vs 2019 levels						
	Q4 25	Q1 26	Q2 26	May'26	Jun'26	Jul'26	Aug'26	Q4 25	Q1 26	Q2 26	May'26	Jun'26	Jul'26	Aug'26
Safran	5.4%	2.7%	-1.6%	-1.6%	-1.0%	1.5%	2.2%	18.0%	20.2%	17.9%	19.3%	15.4%	18.7%	19.0%
MTU	5.5%	2.0%	-1.5%	-2.0%	-0.9%	3.0%	3.9%	9.4%	14.8%	9.8%	9.7%	7.7%	12.9%	13.7%
Rolls-Royce	7.8%	4.5%	2.4%	2.2%	2.8%	2.8%	3.1%	13.1%	19.1%	13.7%	14.9%	11.1%	11.7%	12.6%
GE	5.0%	1.4%	-1.9%	-1.9%	-0.9%	1.2%	1.6%	13.6%	14.2%	12.9%	14.1%	10.8%	13.2%	13.5%
Pratt	6.3%	3.3%	0.2%	0.0%	0.9%	4.5%	5.6%	13.5%	20.8%	15.7%	16.0%	13.0%	17.7%	18.8%
Melrose	3.3%	-0.1%	-3.1%	-3.5%	-2.4%	0.1%	0.6%	1.4%	2.7%	-0.6%	-0.4%	-2.0%	1.0%	1.3%

Source: Cirium, Barclays Research

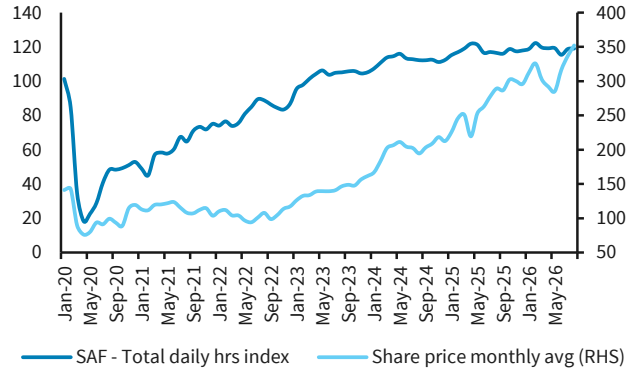
We track the engine utilisation of each engine OEM and try to assess the correlation with the share price movements.

FIGURE 20. Safran daily flight hours YoY growth



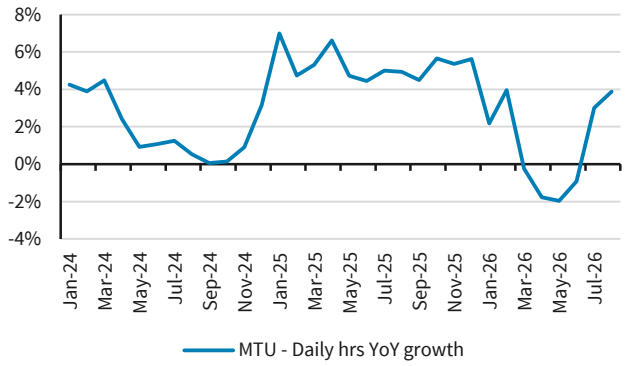
Source: Cirium, Barclays Research

FIGURE 21. Safran daily flight hours (indexed to 100 in the same month of 2019) vs Safran share price



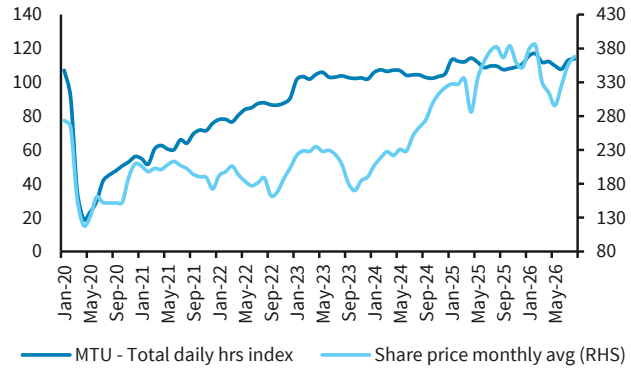
Share price in EUR
Source: Cirium, Bloomberg, Barclays Research

FIGURE 22. MTU daily flight hours YoY growth



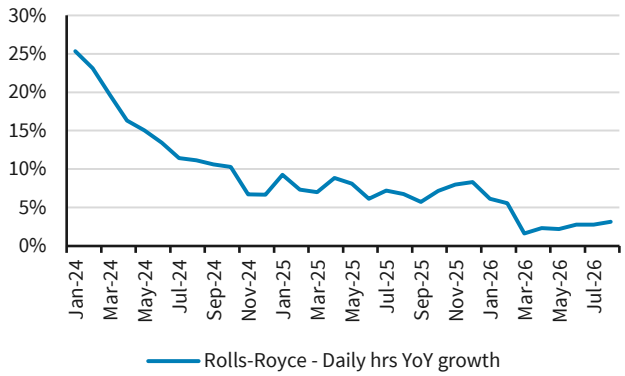
Source: Cirium, Barclays Research

FIGURE 23. MTU daily flight hours (indexed to 100 in the same month of 2019) vs MTU share price



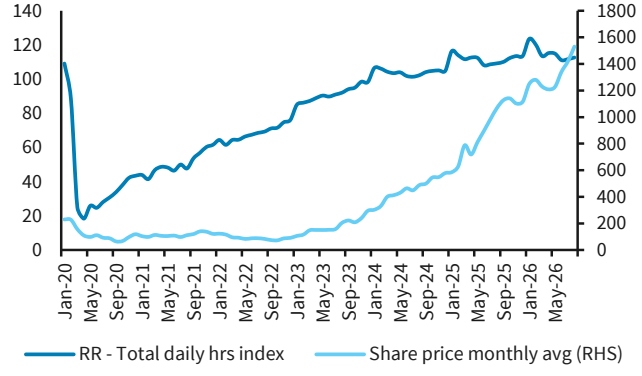
Share price in EUR
Source: Cirium, Bloomberg, Barclays Research

FIGURE 24. Rolls-Royce daily flight hours YoY growth



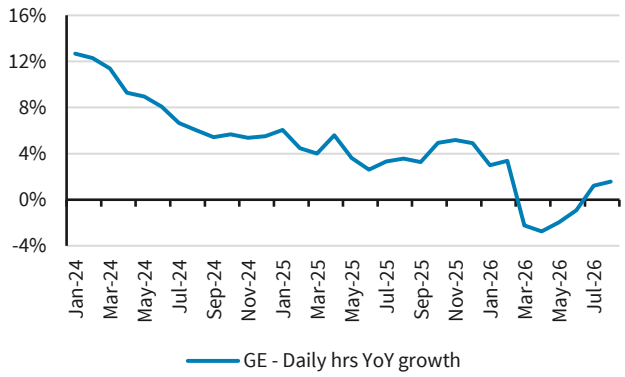
Source: Cirium, Barclays Research

FIGURE 25. Rolls daily flight hours (indexed to 100 in the same month of 2019) vs RR share price



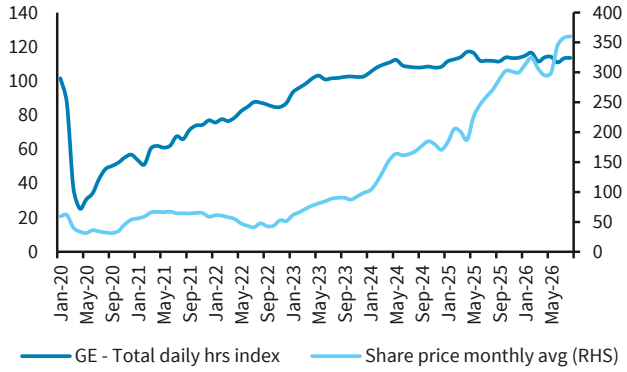
Share price in GBp
Source: Cirium, Barclays Research

FIGURE 26. GE daily flight hours YoY growth



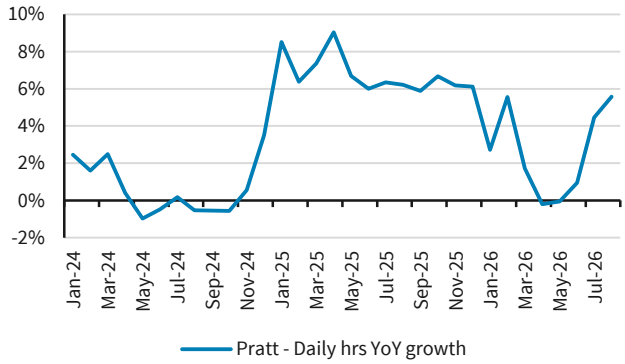
Source: Cirium, Barclays Research

FIGURE 27. GE daily flight hours (indexed to 100 in the same month of 2019) vs GE share price



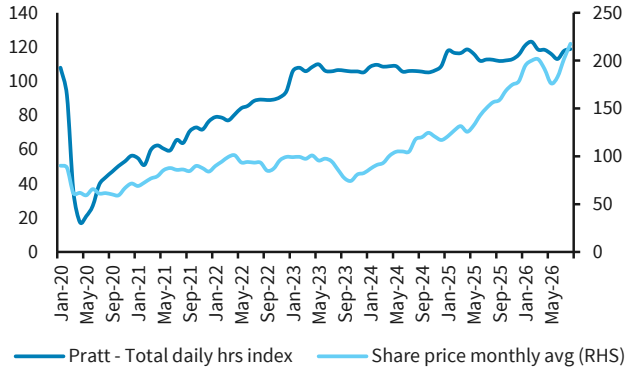
Share price in USD
Source: Cirium, Barclays Research

FIGURE 28. Pratt daily flight hours YoY growth



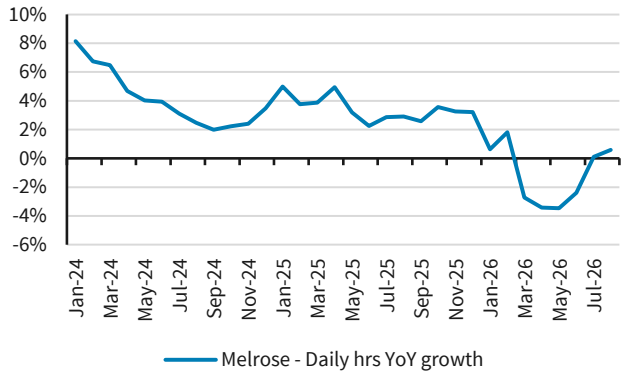
Source: Cirium, Barclays Research

FIGURE 29. Pratt daily flight hours (indexed to 100 in the same month of 2019) vs share price



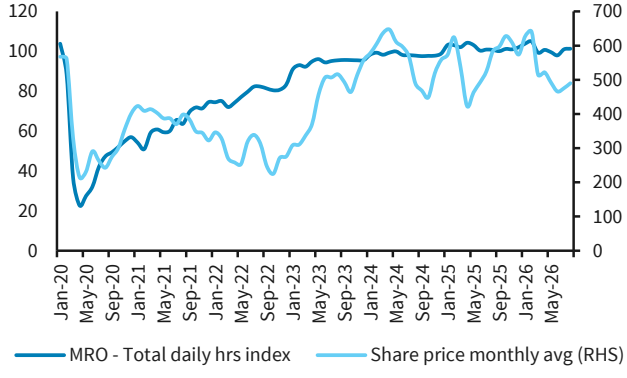
Share price in USD
Source: Cirium, Barclays Research

FIGURE 30. Melrose daily flight hours YoY growth



Source: Cirium, Barclays Research

FIGURE 31. Melrose daily flight hours (indexed to 100 in the same month of 2019) vs Melrose share price

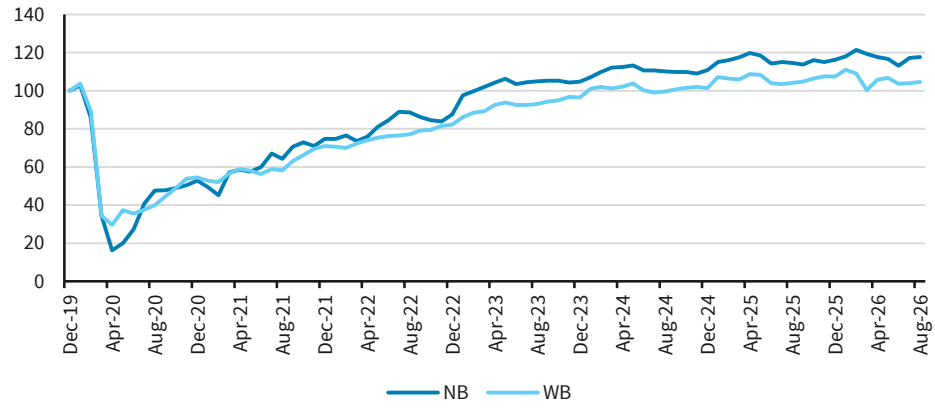


Share price in GBp
Source: Cirium, Barclays Research

Tracking flight hours by aircraft category

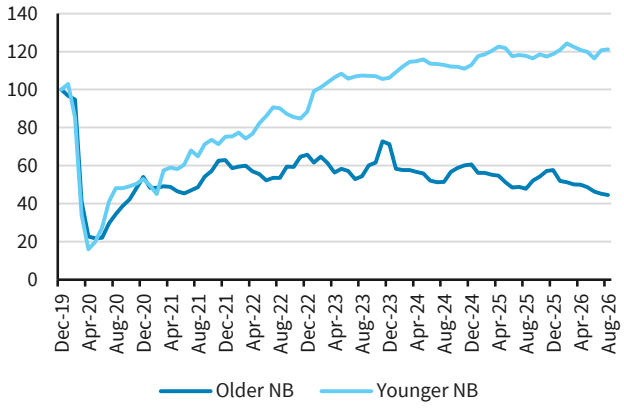
We estimate that narrow-body engine flight hours were 2.7% higher y/y in August (vs 1.9% higher y/y in July and 1.0% lower y/y in June) leading to a 17.6% increase over August 2019 levels. The flight hour growth was mainly driven by the LEAP and GTF engines, which saw +20.8% and +27.1% y/y growth, respectively. Growth in the new-generation NB engine flight hours has mainly been driven by the increase in the installed fleet over the comp period. Wide-body engine flight hours were 0.5% higher y/y in August (vs 0.6% higher y/y in July and 0.2% lower y/y in June) and 4.6% above August 2019 levels. The return to growth in wide-body flight activity reflects the recovery in the Middle East, which serves as a hub for long-haul international travel catered by wide-body aircraft.

FIGURE 32. Total flight hours by aircraft category (indexed to 100 in the same month of 2019)



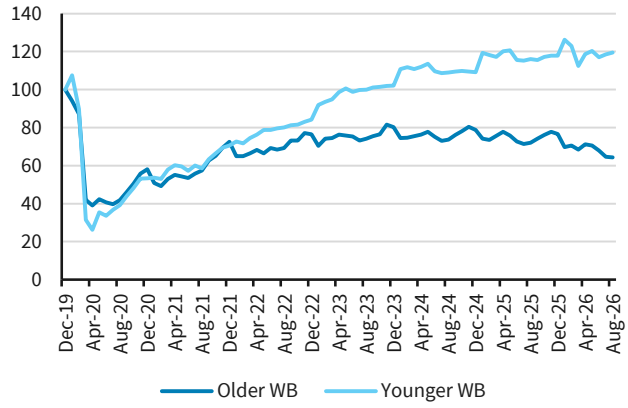
Source: Cirium, Barclays Research

FIGURE 33. Flight hours of older vs newer NB aircraft (indexed to 100 in the same month of 2019)



Source: Cirium, Barclays Research

FIGURE 34. Flight hours of older vs newer WB aircraft (indexed to 100 in the same month of 2019)



Source: Cirium, Barclays Research

FIGURE 35. Flight hour growth by aircraft category

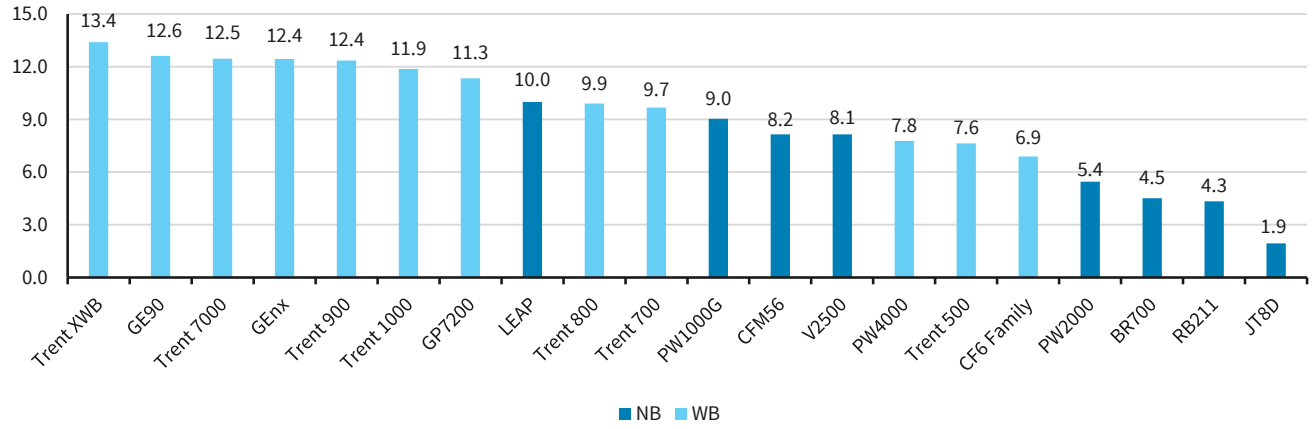
	YoY growth							vs 2019 levels						
	Q4 25	Q1 26	Q2 26	May'26	Jun'26	Jul'26	Aug'26	Q4 25	Q1 26	Q2 26	May'26	Jun'26	Jul'26	Aug'26
NB	5.3%	2.9%	-1.4%	-1.6%	-1.0%	1.9%	2.7%	15.7%	19.6%	15.8%	16.6%	13.1%	17.2%	17.6%
WB	5.4%	0.3%	-1.5%	-1.5%	-0.2%	0.6%	0.5%	7.1%	6.7%	5.3%	6.7%	3.7%	4.0%	4.6%

Source: Cirium, Barclays Research

Among narrow-body engines, LEAP average daily utilisation remained highest at 10.0 hours, followed by the GTF engine at 9.0 hours. The older generation CFM56 and V2500 engines saw average flight activity of 8.2 hours and 8.1 hours, respectively, during the month.

Among wide-body engines, average daily utilisation was highest for the Trent XWB at 13.4 hours followed by the GE90 at 12.6 hours, Trent 7000 at 12.5 hours, GEnx at 12.4 hours, Trent 900 at 12.4 hours and Trent 1000 at 11.9 hours. The Trent 700 powering the older A330 variant saw flight activity of 9.7 hours.

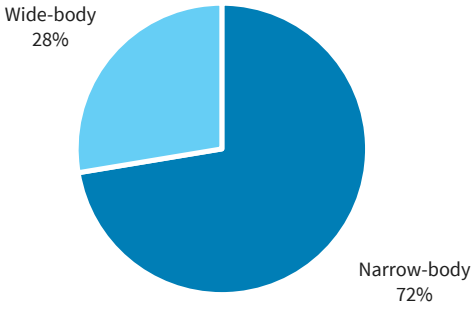
FIGURE 36. Current average of daily utilisation hours



Source: Cirium, Barclays Research

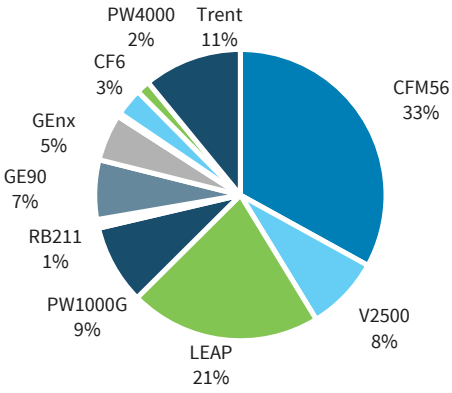
Tracking industry market share

FIGURE 37. Market share of total flight hours by aircraft category



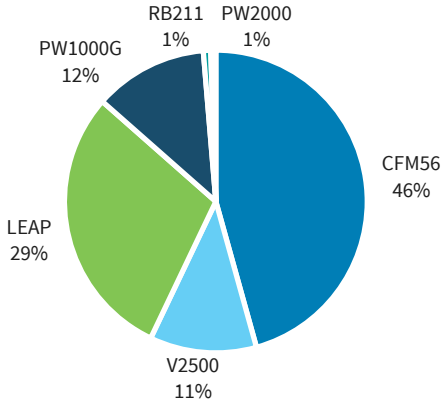
Source: Cirium, Barclays Research

FIGURE 38. Market share of total flight hours by engine type



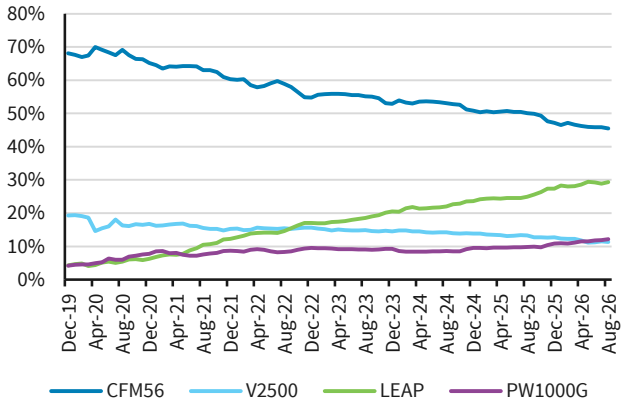
Source: Cirium, Barclays Research

FIGURE 39. Market share of NB flight hours by engine type



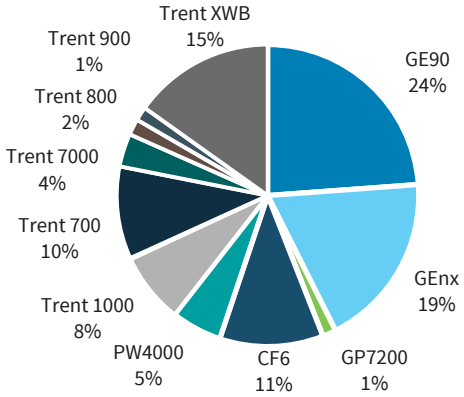
Source: Cirium, Barclays Research

FIGURE 40. Historical market share of NB flight hours



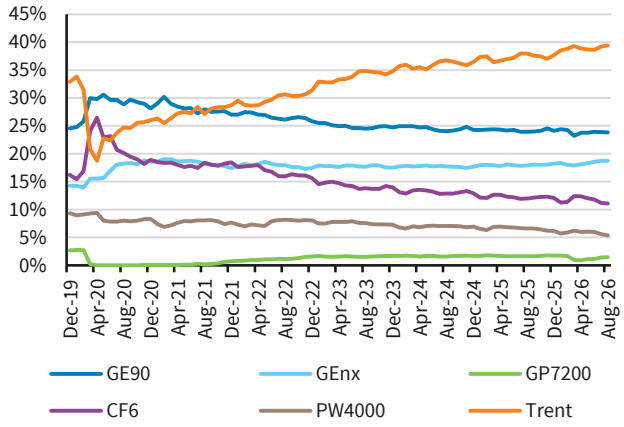
Source: Cirium, Barclays Research

FIGURE 41. Market share of WB flight hours by engine type



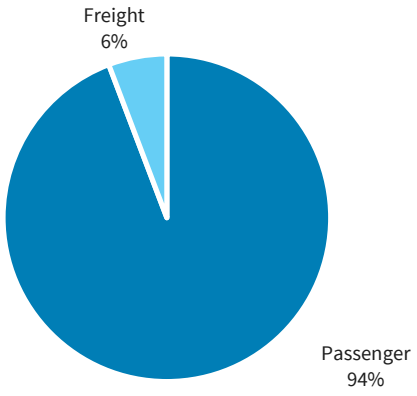
Source: Cirium, Barclays Research

FIGURE 42. Historical market share of WB flight hours



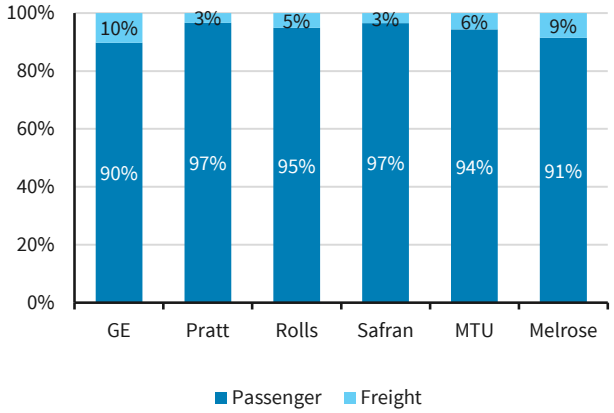
Source: Cirium, Barclays Research

FIGURE 43. Market share of total flight hours by aircraft usage



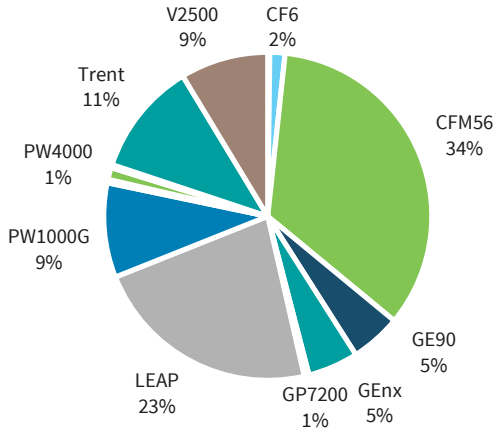
Source: Cirium, Barclays Research

FIGURE 44. Passenger and freighter flight hour share by engine OEM



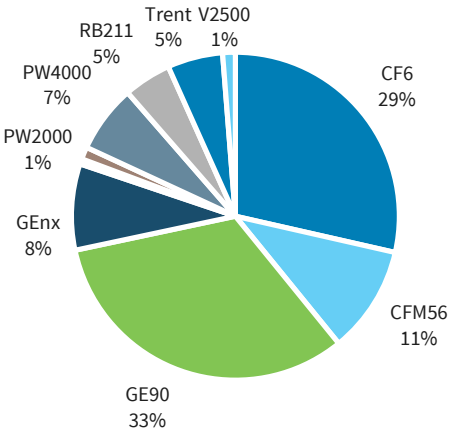
Source: Cirium, Barclays Research

FIGURE 45. Market share of passenger flight hours by engine type



Source: Cirium, Barclays Research

FIGURE 46. Market share of freighter flight hours by engine type

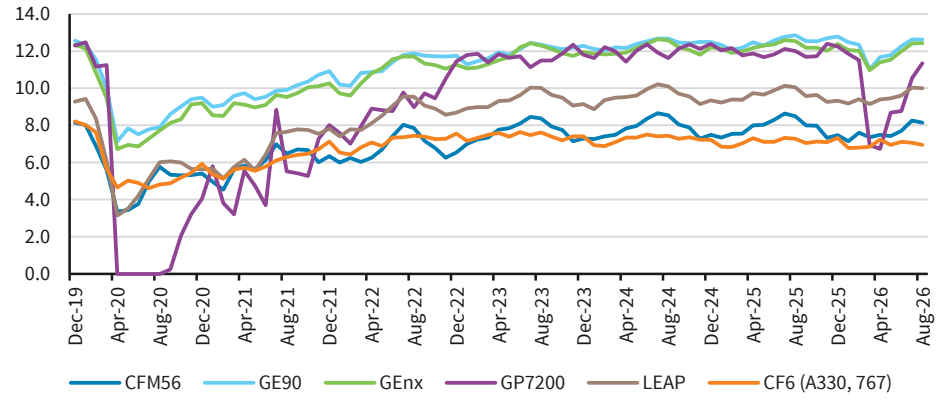


Source: Cirium, Barclays Research

Tracking flight hours by engine type and region

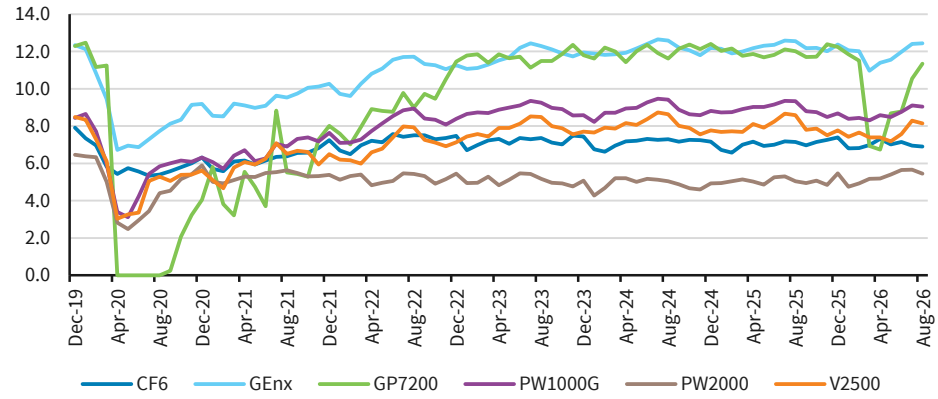
We look at the product portfolio of each engine OEM in order to understand the average daily utilisation trend of each of the engine types.

FIGURE 47. Flight hours of engine platforms on which Safran is a partner (hours/day)



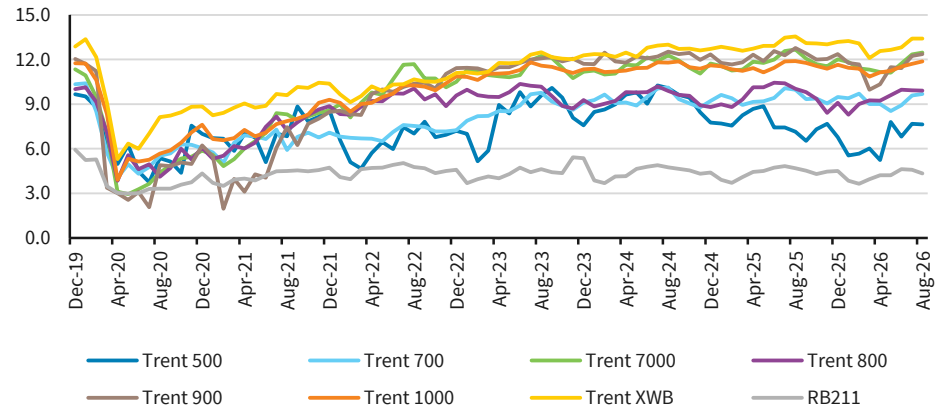
Source: Cirium, Barclays Research

FIGURE 48. Flight hours of engine platforms on which MTU is a partner (hours/day)



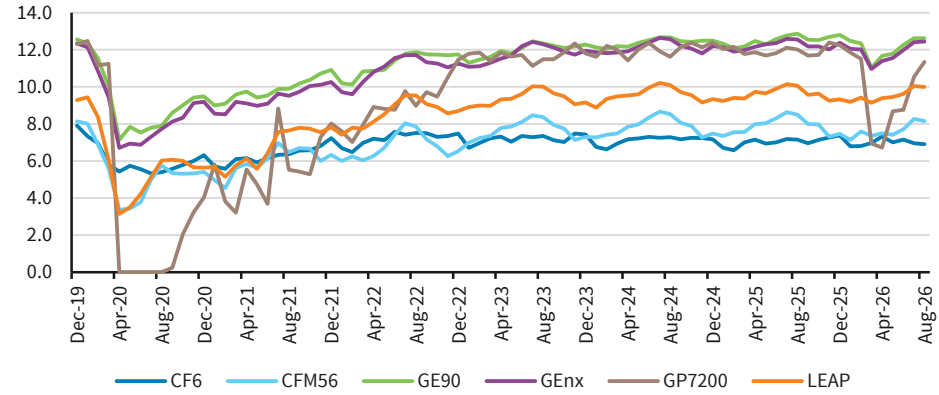
Source: Cirium, Barclays Research

FIGURE 49. Rolls-Royce flight hours by engine type (hours/day)



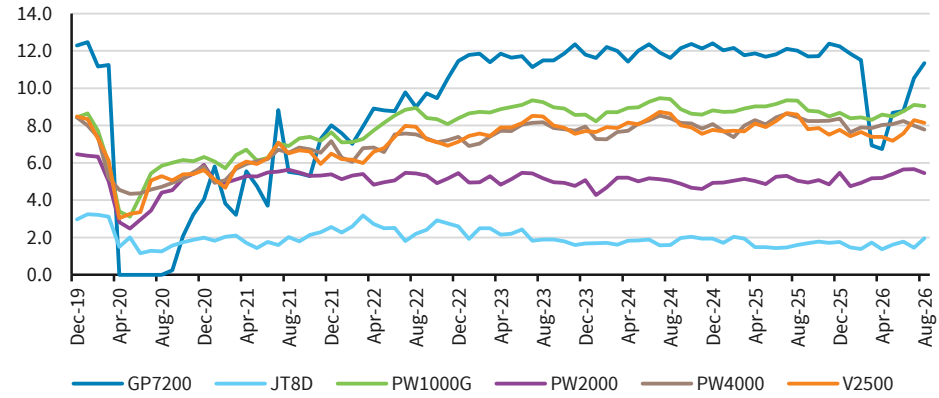
Source: Cirium, Barclays Research

FIGURE 50. **GE flight hours by engine type (hours/day)**



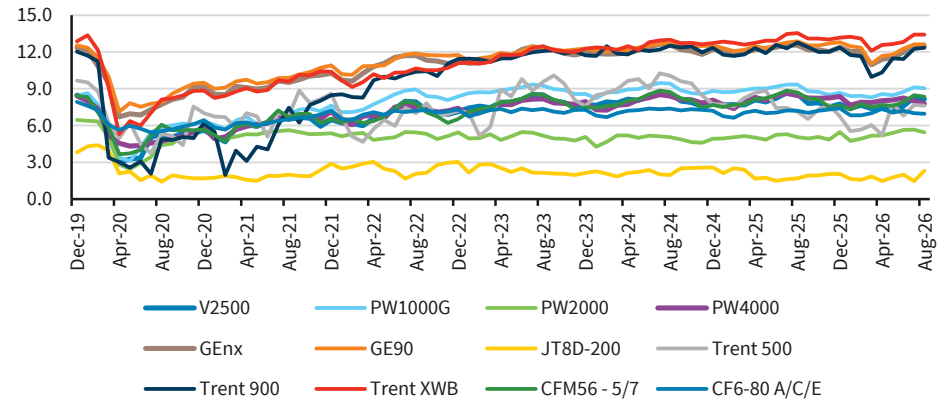
Source: Cirium, Barclays Research

FIGURE 51. **Pratt flight hours by engine type (hours/day)**



Source: Cirium, Barclays Research

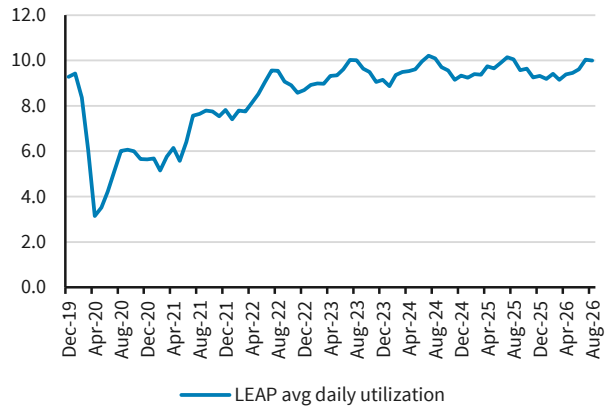
FIGURE 52. **Flight hours of engine platforms on which Melrose is a partner (hours/day)**



Source: Cirium, Barclays Research

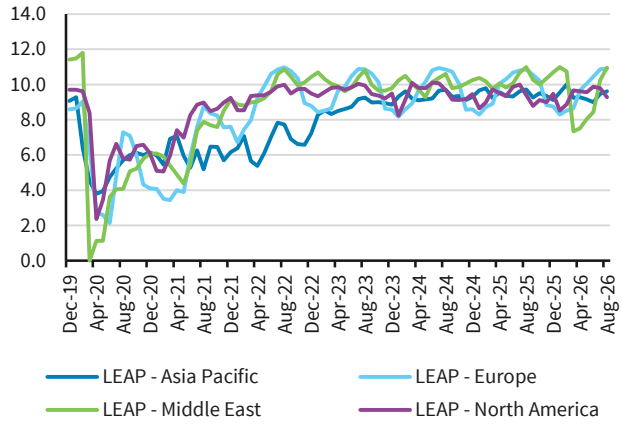
For each of the major engine programs, we look at the utilisation trends in key regions in order to identify the key geographies of demand and flight activity.

FIGURE 53. LEAP engine daily flight activity (hours/day)



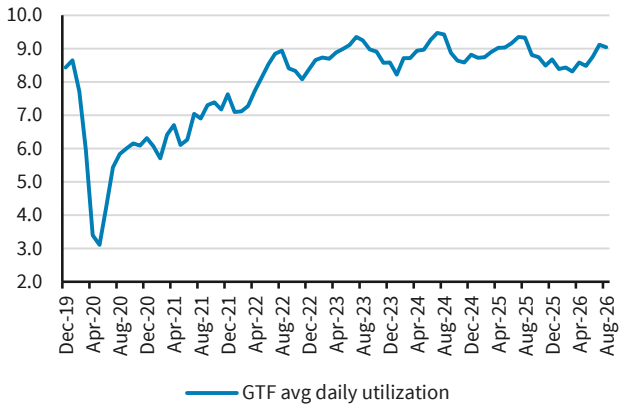
Source: Cirium, Barclays Research

FIGURE 54. LEAP engine daily flight activity by region (hours/day)



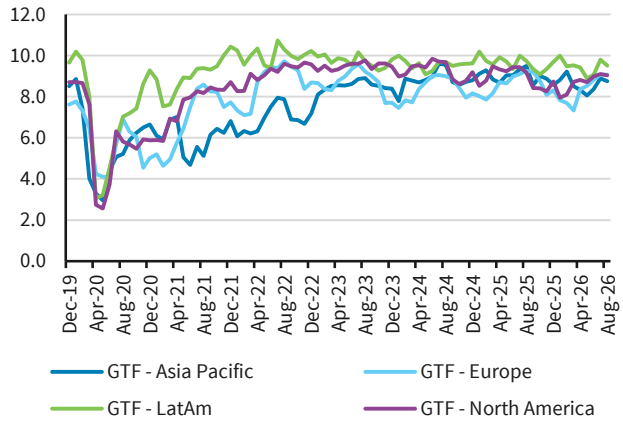
Source: Cirium, Barclays Research

FIGURE 55. GTF engine daily flight activity (hours/day)



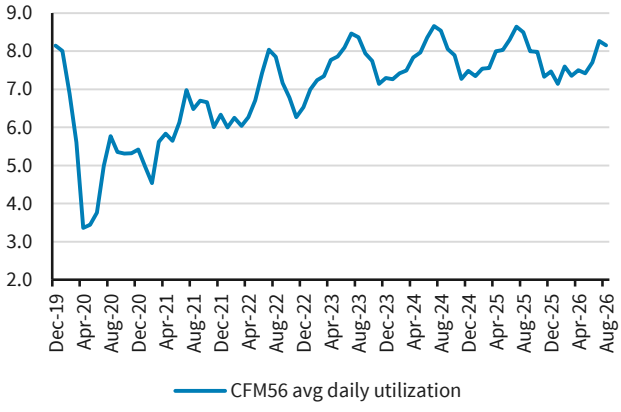
Source: Cirium, Barclays Research

FIGURE 56. GTF engine daily flight activity by region (hours/day)



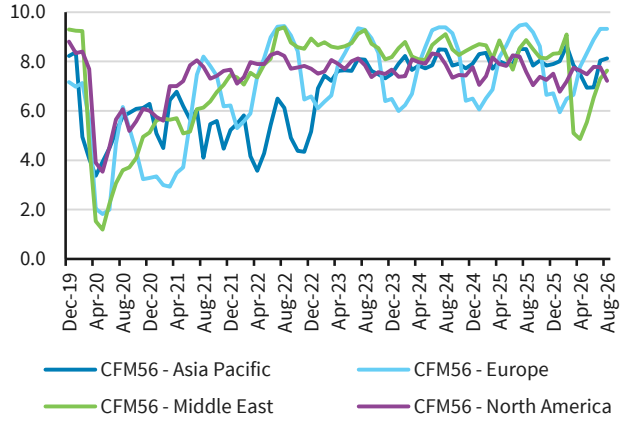
Source: Cirium, Barclays Research

FIGURE 57. CFM56 engine daily flight activity (hours/day)



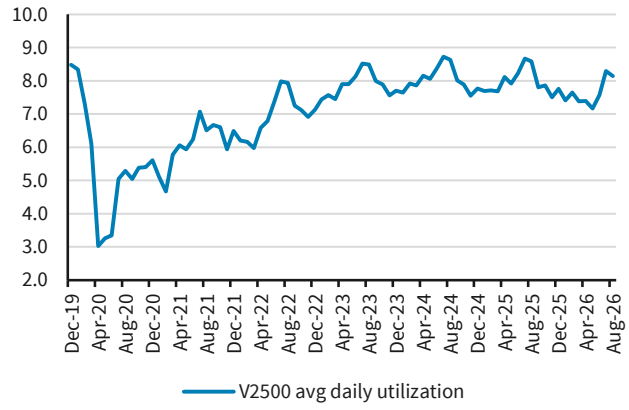
Source: Cirium, Barclays Research

FIGURE 58. CFM56 engine daily flight activity by region (hours/day)



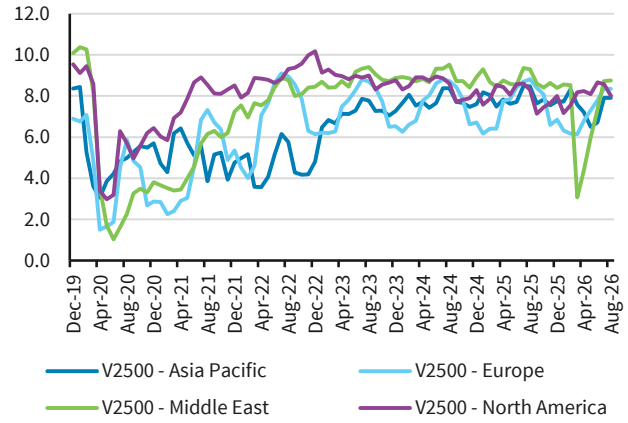
Source: Cirium, Barclays Research

FIGURE 59. V2500 engine daily flight activity (hours/day)



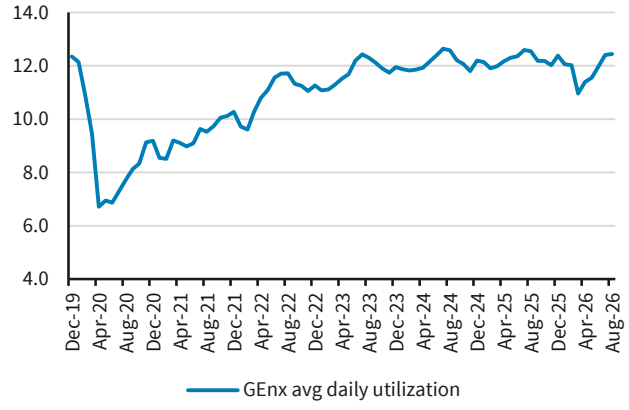
Source: Cirium, Barclays Research

FIGURE 60. V2500 engine daily flight activity by region (hours/day)



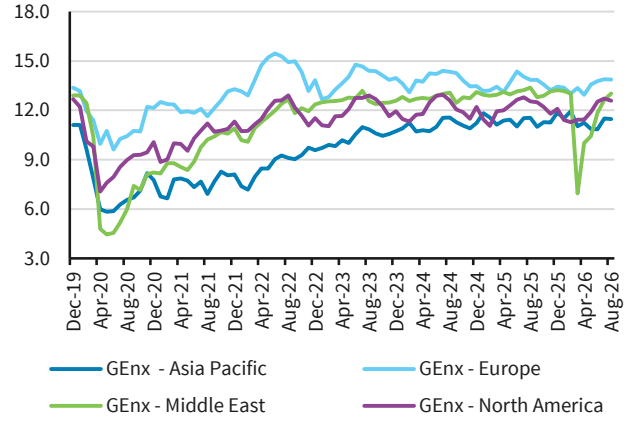
Source: Cirium, Barclays Research

FIGURE 61. GEnx engine daily flight activity (hours/day)



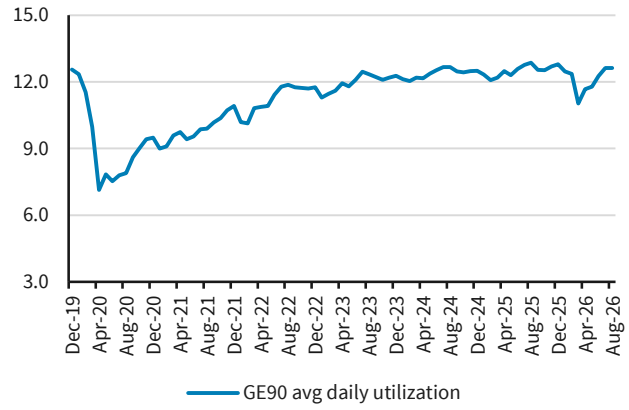
Source: Cirium, Barclays Research

FIGURE 62. GEnx engine daily flight activity by region (hours/day)



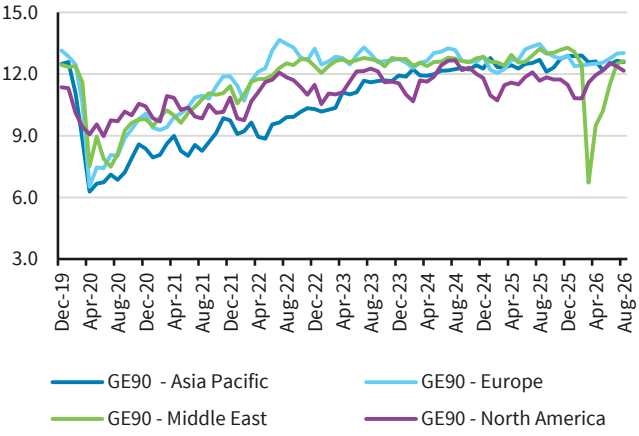
Source: Cirium, Barclays Research

FIGURE 63. GE90 engine daily flight activity (hours/day)



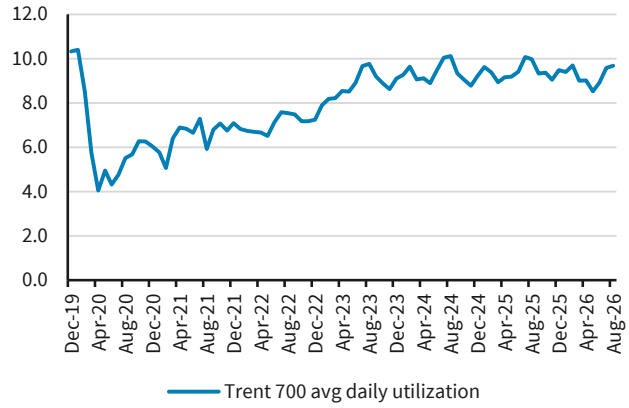
Source: Cirium, Barclays Research

FIGURE 64. GE90 engine daily flight activity by region (hours/day)



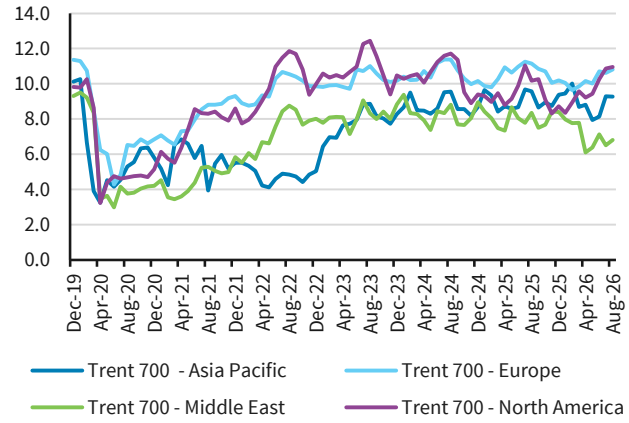
Source: Cirium, Barclays Research

FIGURE 65. Trent 700 engine daily flight activity (hours/day)



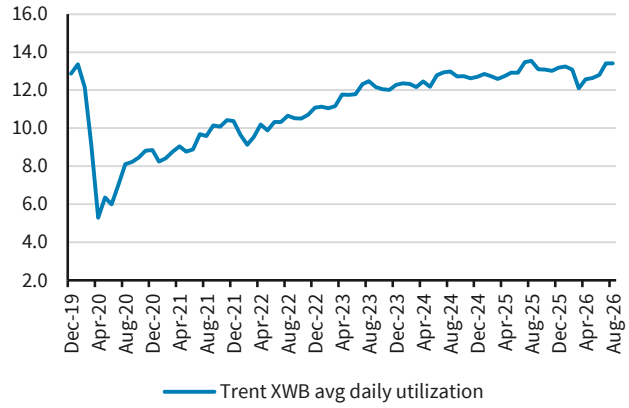
Source: Cirium, Barclays Research

FIGURE 66. Trent 700 engine daily flight activity by region (hours/day)



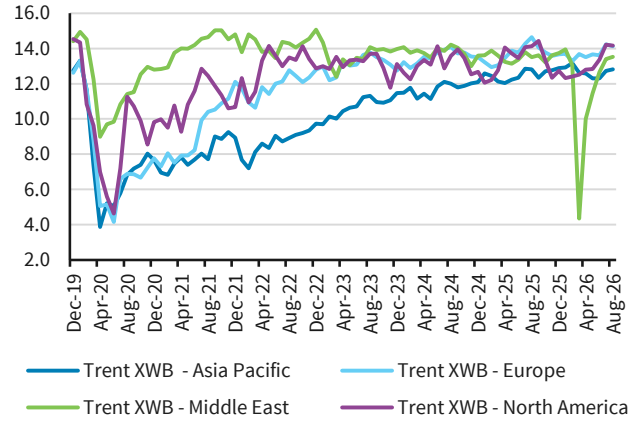
Source: Cirium, Barclays Research

FIGURE 67. Trent XWB engine daily flight activity (hours/day)



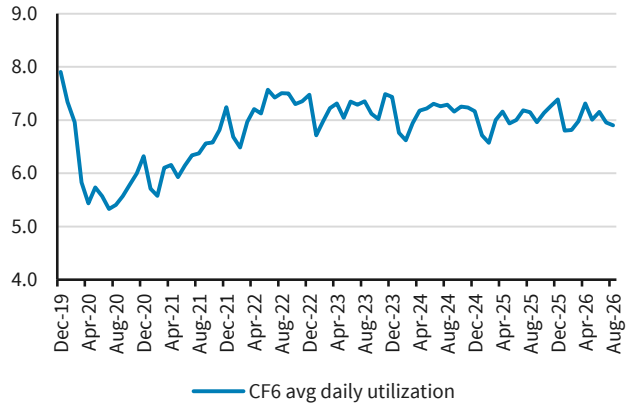
Source: Cirium, Barclays Research

FIGURE 68. Trent XWB engine daily flight activity by region (hours/day)



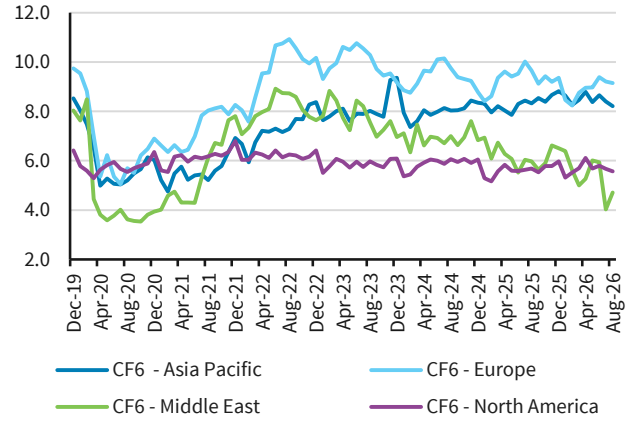
Source: Cirium, Barclays Research

FIGURE 69. CF6 engine daily flight activity (hours/day)



Source: Cirium, Barclays Research

FIGURE 70. CF6 engine daily flight activity by region (hours/day)



Source: Cirium, Barclays Research

APPENDIX – Overview of engine programmes

FIGURE 71. Snapshot of engine programmes powering the various Airbus and Boeing aircraft platforms

	Rolls-Royce	GE	Pratt & Whitney	Engine Alliance	CFM International	IAE
A220			PW1500G			
A320 ceo family					CFM56	V2500
A320 neo family			PW1000G		LEAP-1A	
A330	Trent 700	CF6	PW4000			
A330neo	Trent 7000					
A340 - 200/300					CFM56	
A340 - 500/600	Trent 500					
A350XWB	Trent XWB					
A380	Trent 900			GP7200		
737 NG					CFM56	
737 MAX					LEAP-1B	
747 Classic	RB211 - 524	CF6	PW4000			
747 - 8		GEnx-2B				
757	RB211 - 535		PW2000			
767	RB211 - 524	CF6	PW4000			
777 - 200/200ER/300	Trent 800	GE90	PW4000			
777 - 200LR /300ER		GE90				
777X		GE9X				
787	Trent 1000	GEnx-1B				

Source: Company data, Barclays Research

FIGURE 72. **Safran key commercial, helicopter and military engine programmes**

		Program share	Aircraft Application
Business jet	**1,000 to 115,000 pounds of thrust**		
	Passport	7%	Bombardier Global 7000, 7500 and 8000 business jet
Regional jet	**8,000 and 18,000 pounds of thrust**		
	CF34-10	9.5%	Embraer 190, COMAC C909 (ARJ 21) regional jets
Narrowbody market	**18,500 and 50,000 pounds of thrust**		
	CFM56	50%	B737, A320ceo family
	LEAP	50%	737 MAX, A320neo family, C919
Widebody market	**over 50,000 pounds of thrust**		
	CF6	10-19.7%	A330, 767, LM6000 gas turbine
	GE90	23.7%	B777
	GEnx	7.3%	B787, B747-8
	GE9X	11.2%	B777X
	GP7200	17.5%	A380
Military			
	M53	Full engine	Mirage 2000 fighter jet
	M88	Full engine	Rafale fighter jet
	Atar	Full engine	Mirage F1 fighter jet
	TP400		A400M
	Adour		BAE Systems Hawk trainer aircraft
	Larzac		Alpha Jet training aircraft
Helicopter	**Light helicopters**		
	Arrius		H120, H135 (Airbus), Ka-226T (Kamov), Bell 505 Jet Ranger X, Fennec (Airbus military), A109 LUH (Leonardo)
	Arriel		H125, H130, H145, H155, AS365 N3+, H145M, AS565MBe (Airbus), S-76 C++ (Sikorsky), AC311A-AC312 (AVIC), LCH and LAH (KAI)
	Arrano		Airbus H160
	Medium-weight helicopters		
	MTR390		Tiger
	TM333		Hindustan Aeronautics Ltd's Dhruv
	Ardiden		
	Heavy-lift helicopters		
	Makila		Super Puma family
	RTM322		NH90, AW101 Merlin, Apache UK fighter

Source: Company data, Barclays Research

FIGURE 73. **MTU key commercial and military engine programmes**

	MTU program share	Aircraft Application
Business jet		
PW300	**15-25%**	1000, Dassault Falcon 7X, Cessna Sovereign/ Learjet 60, Do328 JET, Gulfstream G200, Hawker
PW500	**25%**	Cessna Bravo, Cessna Excel
PW800	**15%**	Gulfstream G400, G500, G600
Regional jet		
PW1200G	**15%**	Mitsubishi SpaceJet
PW1500G	**15%**	A220
PW1700/1900G	**17%**	Embraer E-Jet Gen 2
Narrowbody market		
PW2000	**21%**	B757
V2500	**16%**	A320ceo family, Boeing MD-90, KC-390
PW1100G	**18%**	A320neo family
PW1400G	**18%**	MC-21
Widebody market		
GP7000	**23%**	A380
GEnx on 787	**6.6%**	B787-8, B787-9, B787-10
GEnx on 747-8	**6.6%**	B747-8F (Freighter Aircraft), B747-8I (Passenger Aircraft)
GE9X	**4%**	B777X
CF6	**9%**	A310, A330, B767, B747-400, MD-11, DC-10
Military Aircraft		
EJ200	**30%**	Eurofighter
EJ199	**40%**	Tornado
F414	**4%**	F414: F/A-18 E/F Super Hornet; EA-18G Growler
Military Tranport		
TP400	**28%**	A400M
Military Helicopter		
MTR390	**31-40%**	Tiger
T408	**18.4%**	CH-53K

Source: Company data, Barclays Research

FIGURE 74. Rolls-Royce key commercial, helicopter and military engine programmes

		Partner	Aircraft Application
Business jets			
	Pearl 700		Gulfstream G700, G800
	Pearl 15		Bombardier Global 5500, Global 6500
	Pearl 10X		Dassault Falcon 10X
	AE 3007		Cessna Citation X and X+ jets, Embraer Legacy 600 and Legacy 650 jets
	BR710		Bombardier Global 5000, Global 6000 jets, Gulfstream G500, G550 jets
	BR725		Gulfstream G650
	TAY		Gulfstream G350, G450
Regional jets			
	AE 3007		Embraer ERJ 135, E140, E145
	TAY		Fokker 70, Fokker 100
Narrowbody market			
	RB211 - 535		757
Widebody market			
	Trent 500		A340 - 500/600
	Trent 700		A330ceo
	Trent 7000		A330neo
	Trent 800		777 - 200/200ER/300
	Trent 900		A380
	Trent 1000		787
	Trent XWB		A350
	RB211 - 524		747, 767
Military : Combat jets			
	EJ200	Avio, ITP, MTU	Eurofighter Typhoon
	RB199	MTU, Avio	Panavia Tornado
	Pegasus		Harrier aircraft
	Adour		BAE Systems Hawk trainer aircraft, European Unmanned Combat Air Vehicle (UCAV)
	Spey		Gulfstream GII and GIII, AMX Strike aircraft
	F130		B-52 Bomber, C-37, E-11 BACN aircraft
Military : Transport, Patrol, Tactical aircraft			
	TP400	MTU, Safran, ITP	Airbus A400M
	AE 2100		Lockheed Martin C-130J Hercules, LM-100J, Alenia C-27J, Saab 2000, ShinMaywa US-2
	T56		C-130, NOAA Hurricane Hunter
	TAY		Gulfstream IV-SP aircraft
	BR710		Gulfstream V, Bombardier Global Express
	M250 Turboprop		Light fixed-wing aircraft
Helicopter			
	AE 1107C		V-22 tiltrotor, Global Hawk, C-130J aircraft
	MTR 390	MTU, Turbomeca	Airbus Tiger Helicopter
	CTS800	Honeywell	AgustaWestland Super Lynx and Wildcat, Sikorsky X2, TAI T625, T129 ATAK
	M250 Turboshaft		Bell 206, MD 500, Sikorsky S-76
	Gnome		AgustaWestland Sea King
	Gem		AgustaWestland Lynx

Source: Company data, Barclays Research

FIGURE 75. GE key commercial, helicopter and military engine programmes

		Partner	Aircraft Application
Widebody market			
	GEnX		787, 747-8
	GE90		777
	GE9X		777X
	CF6		747, 767, A300, A310, A330 (early variants)
Narrowbody market			
	LEAP	Safran	A320neo, 737 MAX, COMAC C919
	CFM56	Safran	A320ceo, 737 Classic (300/400/500 model), 737 NG (600/700/800/900 model)
Regional & Business Jets			
	Catalyst		Beechcraft Denali, Eurodrone, XTI TriFan 600
	CT7		Saab 340, Let L-610G
	CF34		Embraer 190/195, Embraer Lineage 1000, Embraer E170/175, ARJ21, Bombardier CRJ100/200/440, Bombardier CRJ700/900/1000, Bombardier Challenger 604/605/650
	HF120	Honda	Honda HA-420 HondaJet, Spectrum S-40 Freedom, Cessna CitationJet CJ1
	H-Series (H75/H80/H85)		Diamond Dart 550, Dornier Do-28 G92, Nextant G90XT, Thrush 510G, Technoavia Rysachok, Let L-410 UVP-E20, Let L-410NG, CAIGA Primus 150
	Passport		Bombardier Global 7500 and 8000
Military/Defence			
	CF6		KC-10 Extender, E-4B NAOC
	CT7		CASA/IPTN CN-235
	F108		KC-135R Stratotanker, RC-135, E-6B Mercury
	F110		F-14 Tomcat, F-15EX Eagle II, F-16 Fighting Falcon
	F138		Lockheed C-5M Super Galaxy
	F404		F/A-18 Hornet, F-117 Nighthawk, Saab JAS39 Gripen, T-7A Red Hawk, KAI T-50/TA-50/FA-50, HAL Tejas Mk1, TAI Hurjet, F-20 Tigershark
	F414		F/A-18E/F Super Hornet, Boeing E/A-18G Growler, Saab JAS 39E/F Gripen NG, KAI KF-21 Boramae, HAL Tejas Mk2, NASA X-59 QueSST
	XA100 (Prototype)		F-35A, F-35C
Helicopters			
	CT7		Bell 214ST / 525, Sikorsky S-70 / S-92, AgustaWestland AW101, AgustaWestland AW189, KAI KUH-1 Surion, Sikorsky UH-60 Black Hawk, Sikorsky SH-60 Seahawk, Bell AH-1W SuperCobra, Bell AH-1Z Viper, Bell UH-1Y Venom
	T408		CH-53K King Stallion helicopter
	T700		Boeing AH-64 D/E, Bell AH-1Z Viper, Bell UH-1Y Venom, Sikorsky UH-60L/M and S70 Black Hawk, Sikorsky MH-60R/S Sea Hawk, Sikorsky HH-60W Jolly Green II, Sikorsky VH-60N White Hawk, Sikorsky MH-60T Jayhawk, Sikorsky S-70i Firehawk, KAI Surion
	T901		Sikosrsky Black Hawk (ITEP program), Boeing AH-64 Apache

Source: Company data, Barclays Research

FIGURE 76. Pratt & Whitney key commercial, helicopter and military engine programmes

		Aircraft Application
Commercial		
	GTF	A320neo, A220, Embraer E-Jets E2
	V2500	A320ceo
	PW4000-94	747, 767, MD-11, A300, A310
	PW4000-100	A330-300, A330-200, A330-200/300
	PW4000-112	777-200, 777-200ER, 777-300
	PW2000	757
	JT8D	727, 737-100/-200, MD-80, DC-9, Aerospatiale SE210 Caravelle Super 10 Thru 12, Dassault-Mercure Breguet, Joint Surveillance Target And Radar System (JSTARS), Kawasaki C-1, Super 27 Re-Engining Program
	JT9D	747, 767, A300, A310, DC-10
Regional		
	PW127XT	ATR 42 series, ATR 72 series, D328eco
	PW100/150	Embraer EMB120 Brasilia, ATR 42-300/320, De Havilland Q100, De Havilland Q300, De Havilland CL-215T/CL-415
Business		
	PW800	Dassault Falcon 6X, Gulfstream G500, Gulfstream G600
	PW600	Eclipse 500, Cessna Mustang, Embraer Phenom 100
	PW500	Cessna Citation Bravo, Cessna Citation Ultra Encore, Cessna UC-35 C/D, Cessna Citation Encore+, Embraer Phenom 300, Cessna Citation Excel, Cessna Citation XLS, Cessna Citation XLS+
	PW300	Bombardier Learjet Model 60, Gulfstream G200, Dornier 328JET, Cessna Citation Sovereign, Cessna Citation Sovereign+, Cessna Citation Latitude, Dassault Falcon 7X, Dassault Falcon 8X, Hawker 4000, Dassault Falcon 2000EX/DX
General		
	PT6E-66XT	Daher TBM 960
	PT6E-67XP	Pilatus PC-12 NGX
	PT6A	Cessna 208/208B Caravan, Beechcraft King Air 200/250/350 Series, Pilatus PC-12 / PC-12 NG / PC-21, Air Tractor AT-802 / 502 / 402 Series, Thrush 510P / 550P / 710P, Viking DHC-6 Twin Otter 400, Quest Kodiak 100, Embraer EMB-312 Tucano, EMB-314 Super Tucano, Cessna SkyCourier
Helicopters		
	PW200	Airbus Helicopters (EC135P1, H135, H135P3), ANSAT-CIV, ANSAT-U, Bell (427, 429), Leonardo (A109 Grand, A109E Power, AW109 Nexus), MD Explorer (900, 902), PW207V
	PW210	Leonardo AW169, Sikorsky S-76D
	PT6C	Leonardo AW609, AW139, TEMSCO UH-1 Global Eagle, Airbus Helicopters H175
	PT6T	Leonardo Helicopters (AB212, AB412), Sikorksy S-58T, Bell (Model 212, 412/SP/HP/EP/E/EPI, CFUTTH CH-146 Griffon), Sikorsky S-58T
	PT6B	Sikorsky S-76B, Leonardo A119 Koala
Military		
	XA103 (in development)	
	F135	F-35 Lightning II
	F119	F-22 Raptor
	F117	C-17 Globemaster III
	F100	F-15, F-15EX, F-16
	PW4062/F139	KC-46 Pegasus
	TF33	B-52 bomber, E-3 AWACS
	V2500	Embraer C-390 Millenium

Source: Company data, Barclays Research

FIGURE 77. Melrose key commercial and military engine programmes

		Engine share on program	Aircraft Application
Regional jet			
	AE 3007	**long-term agreement**	Embraer ERJ 135, E140, E145
	CF34	**long-term agreement**	Embraer 190, COMAC C909 (ARJ 21) regional jets
	PW1500G	**4-7%**	A220
	PW1900	**4-7%**	Embraer E-Jet Gen 2
Narrowbody market			
	PW2000	**4-7%**	B757
	V2500	**2-5%**	A320ceo family, MD-90, KC-390
	PW1100G	**4-7%**	A320neo family
	CFM56-5/7	**<2%**	B737, A320ceo family
	LEAP-1B	**long-term agreement**	B737 MAX
Widebody market			
	Trent 700	**long-term agreement**	A330ceo
	Trent 7000	**long-term agreement**	A330neo
	GP7000	**long-term agreement**	A380
	Trent 1000	**long-term agreement**	B787
	PW4000-94/100	**long-term agreement**	B747-400, 767-200/-300
	PW4000-112	**4-7%**	B777-200/-300
	Trent 900	**2-4%**	A380
	Trent 500	**2-4%**	A340 - 500/600
	Trent XWB-84K	**2-4%**	A350 - 900
	GEnx	**2-5%**	B787, B747
	CF6	**2-4%**	A310, A330, B767
	GE90	**2-4%**	B777
Military Aircraft			
	RM12 and from 2025 RM16		Gripen fighter jet
	F135	**long-term agreement**	F35

Source: Company data, Barclays Research

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